FWA-02 / HEALTHCARE FWA KNOWLEDGE BASE

Provider Risk

Explainable prioritization and evidence-led review

Risk policy and governance

Scope

This synthetic knowledge base describes how a fictional payer prioritizes provider reviews. It contains no real DOH data or allegations. Provider identifiers, thresholds, peers and cases are invented. Risk is the likelihood that a case merits review under these teaching rules; it is not a probability that a provider committed fraud.

PR-P01 | Comparable peers

Compare providers within a documented specialty, facility type, service mix and observation window. Apply minimum activity criteria before showing a percentile. In this exercise a stable peer comparison requires at least 200 accepted lines for the provider and at least 10 providers in the peer group. Below these thresholds, label the result insufficient comparison data rather than assigning a low score.

PR-P02 | Explainability

Every displayed score must expose its component features, observation window, source refresh time and model or rule version. Missing inputs are not zero. A provider with unavailable identity linkage should be routed to data-quality review, not automatically treated as safe or high risk. Demographic or geographic proxies must not substitute for case evidence.

PR-P03 | Human accountability

The score routes a review queue only. A specialist checks evidence before a case decision; an independent reviewer handles disputes. Record what the evidence supports, including legitimate explanations. Do not publish a list of suspected fraudulent providers from this training output. The exercise measures review quality and workload as well as detections.

RAG objective

Learn metadata filtering and retrieval across multiple documents. A question about PRV00092 may need claim facts from FWA-01 and interpretation rules from this PDF. Preserve provenance so the assistant does not present an illustrative score as a clinical or legal conclusion.

Risk workflow and illustrative score

Preparation

Aggregate provider facts over the same 30-day window used in FWA-01. Reconcile provider identifiers and facility relationships. Calculate features only from records available at the scoring cutoff; later case outcomes would leak future information. Preserve missingness indicators and the exact peer cohort used for each feature.

Illustrative scoring rule

Component	Weight and synthetic PRV00092 input
Duplicate candidate percentile	0.30 x 90 = 27.0
Coding-intensity percentile	0.25 x 80 = 20.0
Unusual utilization percentile	0.20 x 70 = 14.0
Documentation-gap percentile	0.15 x 60 = 9.0
Unresolved-case percentile	0.10 x 50 = 5.0
Weighted review score	75.0 out of 100

Meaning and constraints

The percentiles above are invented normalized feature values; they are not percentages of fraudulent claims. The weighted sum is 75.0. For this exercise, scores below 40 use routine monitoring, 40 to below 70 use analyst screening, and 70 or more use priority review. These boundaries are training choices and have no regulatory standing. Correlated components may double-count similar behavior.

Queue and resolution

Create one provider review record containing score version PR-R1, feature window, top reasons, related claims and an assigned reviewer. The reviewer checks source quality, case mix and legitimate operational changes. Resolve as no concern identified, data correction, focused audit or investigation referral. Keep closed cases visible to evaluation without repeatedly re-opening them from unchanged alerts.

Outcome monitoring

Monitor review yield by specialty and volume, median days to resolution and appeal overturn rate. A high yield may reflect selective review; it does not prove strong recall. Estimate missed issues using a separately designed random review sample. Compare performance across relevant cohorts and inspect reasons for disparities before changing thresholds.

Metrics, quality checks and scenarios

Synthetic monthly portfolio

The fictional portfolio contains 100 providers, of which 80 meet peer eligibility and 20 lack sufficient comparison data. Among the 80 eligible providers, 15 enter priority review, 25 enter screening and 40 remain in routine monitoring. Eligibility coverage is 80 / 100 = 80%. Priority share among eligible providers is 15 / 80 = 18.75%.

Review outcomes

Ten priority providers have completed review; four require corrective action, four are supported and two need data correction. Corrective-action yield is 4 / 10 = 40%, not 4 / 15. Median resolution time is an illustrative 12 days. Two decisions are appealed and one is overturned; report the counts with the 50% overturn rate because the denominator is too small for a stable comparison.

PR-S01 | New clinic

A new clinic has 80 accepted lines and a high apparent utilization ratio. Expected handling: mark insufficient comparison data under PR-P01 and use manual screening if there is other evidence. Do not extrapolate a stable percentile from a tiny denominator. Collect a longer window while keeping service dates and policy periods explicit.

PR-S02 | Specialty drift

PRV00092 adds a legitimate rehabilitation service. Average visits per member rise, but a general-clinic peer cohort has not been updated. The analyst checks the service mix and recalculates within an appropriate cohort. Retain both score versions and document the change; do not silently overwrite the original review evidence.

PR-S03 | Shared address

Three synthetic providers share a building and billing vendor. A network view shows close connections. This may be ordinary co-location, not collusion. Request ownership and contractual context before escalation. A graph edge should store relationship type, source and confidence; sharing a telephone number does not establish common ownership.

Quality controls

Check provider key uniqueness, stale reference records, cohort stability and feature availability before publishing scores. Reconcile the sum of tier counts to the eligible population. Keep score drift separate from behavior drift: a revised rule can change a score even when underlying claims are unchanged.

RAG exercises and interview anchors

Intermediate retrieval task

Build an index over FWA-01 and FWA-02 with doc_id, section_id, page, version and synthetic_flag metadata. Ask: Why does PRV00092 receive priority review, and does that prove fraud? Retrieve the weighted score and PR-P03. The correct answer says the illustrative score is 75 and priority review is a routing decision, not proof.

Metadata filtering task

Ask the same policy question with specialty and effective-date filters. Store those filters outside untrusted user text when they enforce permissions. Compare retrieval with and without the doc_id filter. If a narrow filter removes all relevant evidence, respond with insufficient evidence rather than silently broadening into an unauthorized corpus.

Evaluation task

Create a small question set covering the score calculation, minimum peer population, missing features, provider appeal and new-clinic scenario. Identify the expected evidence page for each question. Measure retrieval hit rate separately from whether the generated answer is correct. Inspect cases where the right source is retrieved but the model misinterprets a denominator.

Interview Q1 | Why not train on every closed case?

Closed cases can contain inconsistent labels, selection bias and information unavailable at prediction time. A strong answer discusses label governance, time-based splits, reviewer agreement and leakage controls. Historical decisions are operational outcomes, not automatically ground truth.

Interview Q2 | What does a percentile mean?

It describes relative position in a defined peer distribution. It is neither an absolute utilization rate nor a calibrated fraud probability. Explain the peer cohort, minimum volume and uncertainty. Interview Q3: Why combine structured features and RAG? Structured computations produce reproducible scores; RAG explains policies and supporting evidence using cited text.

Advanced extension

Add relationship-aware retrieval using provider-to-facility and provider-to-case links. Keep graph evidence separate from inferred suspicion. Evaluate whether the extra context improves answer accuracy enough to justify complexity and latency. Use FWA-06 for reranking, security, deployment gates and observability.